

The PSG-386 Conjecture

by Sven Nilsen, 2026

The Prime Symmetric Goldbach conjecture (PSG) states that for any natural number $n > 1$ that is divisible by 6, there is a 1-or-prime p such that $n = a + p = b - p$ where a, b are primes.

Notice that $a + p$ is almost symmetric, except for the case $p = 1$. This raises the question for which numbers n they can be swapped, such that $c = n + a$ gives $n = p + a = c - a$, where p and a have same values as in the PSG conjecture and c is a prime. As before, p is a 1-or-prime.

By adding an additional constraint that $q = n + a$ is a prime, exhaustive search (up to $6 * 30\,000$ so far) shows that most n have this property, except the numbers of the following sequence of $n / 6$:

8, 23, 76, 87, 89, 93, 96, 116, 163, 179, 181, 274, 362, 381, 386

The PSG-386 Conjecture states that there is no higher natural number than 386, multiplied with 6, that does not have the symmetry that the two lower number in the PSG conjecture can be swapped to find a third prime. However, there might be more than one solution for the PSG conjecture, so the PSG-386 only states that one solution has this symmetry. It does not hold for all solutions.